

US to European Bidding Language Conversion

- S : set of sellers
- B : set of buyers
- $T = \{1, \dots, 24\}$: set of time periods

1 Sellers conversion

For each $s \in S$ we have the following data:

- $\underline{P}_{st}, \bar{P}_{st} \forall t \in T$: min and max production levels per period
- $\underline{R}_s$: min uptime
- h_s : no-load cost
- set of bids: $\{(q_{stl}, c_{stl})\}_{l \in \beta_s^t} \forall t \in T$, where q_{stl} denotes the max. quantity of bid l and c_{stl} is the marginal cost of bid l

Assumptions: $\bar{P}_{st} = \sum_{l \in \beta_s^t} q_{stl}$, $\underline{P}_{st} \leq q_{st1}$, the cost curve is identical in all periods.

1.1 $\underline{R}_s = 0, h_s = 0, \underline{P}_{st} \geq 0$

Create a scalable complex order:

- minimum acceptance volume per hour: $\underline{P}_{st}$

and associated step suborders for the remaining bids.

1.2 $\underline{R}_s > 0, h_s = 0, \underline{P}_{st} \geq 0$

We need to generate all patterns that encode in which period the seller s is committed. There are $O(2^{|T|})$ such combinations.

1.3 Generate patterns

Our problem can be reduced to the following problem:

Consider a segment L of length $|T|$ and a set of segments with lengths in $M = \{1, R_s, R_s + 1, \dots, |T|\}$. Assume there are an infinite number of segments of each length. The goal is to generate all possibilities of obtaining L by concatenating segments with lengths in M . We can do this with dynamic programming.

1.4 Convert patterns to block orders

We create an exclusive group for each $s \in S$. For each pattern i we derive a block order B_{si} .

Example 1.1. Assume seller s has a minimum production level of 2 MWh in each period. The block bid corresponding to the pattern $[1, 1, 12, 1, 1, 8]$ is defined as follows:

- *MAR*: 1
- *limit price*: $c_s^{\min}$
- $H1, H2 = 0$; $Hk = 2$ for $k \in \{3, \dots, 14\}$; $H15, H16 = 0$; $Hk = 2$ for $k \in \{17, \dots, 24\}$
- *block_type* = *exclusive*

This pattern is feasible for any minimum uptime smaller than 13.

For each block order we define linked block orders. Each linked block order has volume 0 for all periods except for one period and $MAR = 0$. The limit price is given by the set of bids.

1.5 $\underline{R}_s > 0, h_s > 0, \underline{P}_{st} \geq 0$

TODO

2 Buyers conversion

For each $b \in B$ we have the following data:

- $\underline{P}_{bt}$: price-inelastic demand per period
- set of price-elastic demand bids $\{(q_{btl}, p_{btl})\}_{l \in \beta_b^t}$

2.1 Inelastic demand

Create a block bid B_b :

- $MAR = 1$
- volumes: $Ht = \underline{P}_{bt} \quad \forall t \in T$
- limit price: a very large number to ensure that the block is always accepted

2.2 Elastic demand

Create step orders.